

Homework 3

Due Wednesday February 18, 2026

Reading:

- Required: Lecture 4-5 Notes
- Optional: Ch. 2 of IPC

Problems:

1. (IPC: 2.01) **Exercise 2.1** As a particular realization of the above example, consider an 8×8 chessboard and a special piece sitting on it. At any time step, the piece will stay still (with probability $1/2$) or move randomly to one of the neighboring positions (with probability $1/2$). Does this process satisfy the detailed balance condition (2.15)? Which positions on the chess board have lower and higher ‘energy’? Compute the partition function.
2. (IPC: 2.02) **Exercise 2.2** *A two-level system.* This is the simplest non-trivial example: $\mathcal{X} = \{1, 2\}$, $E(1) = \epsilon_1$, $E(2) = \epsilon_2$. Without loss of generality, we assume $\epsilon_1 < \epsilon_2$. This example can be used as a mathematical model for many physical systems, such as the spin- $1/2$ particle discussed above.

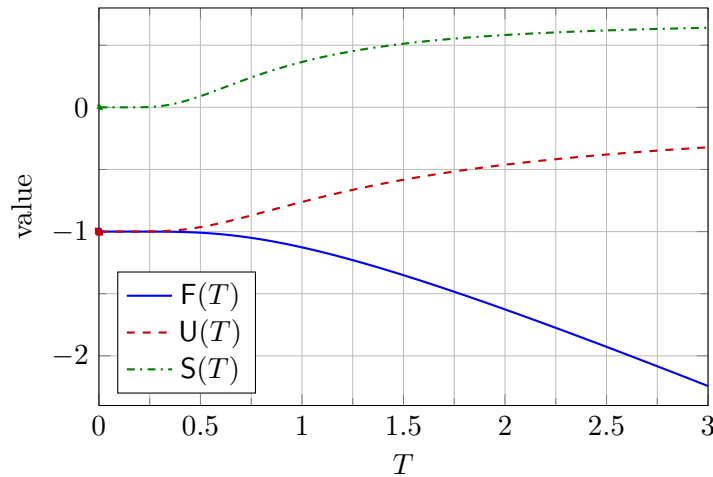
Derive the following for the thermodynamic potentials (where $\Delta = \epsilon_2 - \epsilon_1$ is the energy gap):

$$F(\beta) = \epsilon_1 - \frac{1}{\beta} \log(1 + e^{-\beta\Delta})$$

$$U(\beta) = \epsilon_1 + \frac{e^{-\beta\Delta}}{1 + e^{-\beta\Delta}} \Delta$$

$$S(\beta) = \frac{e^{-\beta\Delta}}{1 + e^{-\beta\Delta}} \beta\Delta + \log(1 + e^{-\beta\Delta})$$

For $\epsilon_1 = -1$ and $\epsilon_2 = 1$, these functions are plotted below (see also Fig. 2.1).



The reader should work out the asymptotics, and verify the general high- and low-temperature behavior shown above.

3. (FGML: 4.01) **Ising Effective field and single-spin conditional.** Consider an Ising model on graph $\mathcal{G}$ with energy

$$E(\sigma) = - \sum_{(i,j) \in \mathcal{E}} J_{ij} \sigma_i \sigma_j - \sum_i h_i^{\text{ext}} \sigma_i, \quad \sigma \in \{\pm 1\}^n.$$

Define $h_i^{\text{eff}} = h_i^{\text{ext}} + \sum_{j \in \partial i} J_{ij} \sigma_j$.

- (a) Derive the conditional distribution $\mathbb{P}(\sigma_i = +1 \mid \sigma_{-i})$ in logistic form.
(b) Show that $\mathbb{E}[\sigma_i \mid \sigma_{\neq i}] = \tanh(\beta h_i^{\text{eff}})$.
4. (FGML: 4.02) **Mean-field self-consistency.** The following fully-connected Ising model on n spins, known as the Curie–Weiss model, is defined by

$$E(\sigma) = -\frac{J}{n} \sum_{1 \leq i < j \leq n} \sigma_i \sigma_j - h \sum_{i=1}^n \sigma_i, \quad \sigma_i \in \{\pm 1\}, \quad J > 0.$$

Let $m = \frac{1}{n} \sum_i \sigma_i$ be the magnetization.

- (a) In this model, one assumes that the magnetization concentrates around its average $m \approx \frac{1}{n} \sum_i \mathbb{E}[\sigma_i]$. Then, this is used to compute new means $\mathbb{E}[\sigma_i \mid \sigma_{-i}]$ based on the other spins. Derive the self-consistency equation for m implied by equilibrium.
(b) For $h = 0$, determine the range of β for which the $m = 0$ fixed point is stable.
5. (IPC: 2.06) **Exercise 2.6** Consider the one-dimensional Ising model in zero field, $B = 0$. Show that when $\delta N < i < j < (1 - \delta)N$, the correlation function $\langle \sigma_i \sigma_j \rangle$ is, in the large- N limit,

$$\langle \sigma_i \sigma_j \rangle = e^{-\frac{|i-j|}{\xi(\beta)}}, \quad \text{where } \xi(\beta) = -\frac{1}{\log(\tanh(\beta))}$$

[Hint: You can either use the general transfer matrix formalism or, more simply, use the identity $e^{\beta \sigma_i \sigma_{i+1}} = \cosh[\beta(1 + \sigma_i \sigma_{i+1} \tanh(\beta))]$.]

Practice Problems (do not hand in):

1. (IPC: 2.03) **Exercise 2.3** We return to the example of the previous section: one water molecule, modeled as a point, in a bottle. We consider the case of a cylindrical bottle of base $B \subset \mathbb{R}^2$ (surface area $|B|$) and height d .

Using the energy function $E(x) = wh(x)$ (see (2.13)), where $h(x)$ is the height of x and w is a constant w , to derive the following explicit expressions for the thermodynamic potentials:

$$\begin{aligned} F(\beta) &= -\frac{1}{\beta} \log(|B|) - \frac{1}{\beta} \log\left(\frac{1 - e^{-\beta wd}}{\beta w}\right) \\ U(\beta) &= \frac{1}{\beta} - \frac{wd}{e^{\beta wd} - 1} \\ S(\beta) &= \log(|Bd|) + 1 - \frac{\beta wd}{e^{\beta wd} - 1} - \log\left(\frac{\beta wd}{1 - e^{-\beta wd}}\right) \end{aligned}$$

Note that the internal-energy formula can be used to compute the average height of the molecule $\langle h(x) \rangle = U(\beta)/w$. This is a consequence of the definition of the energy (see eqn (2.13) and eqn (2.21)). If we plug in the correct constant w , we can find that the average height falls below 49.99% of the height of the bottle $d = 20\text{cm}$ only when the temperature is below 3.2 K.

2. (IPC: 2.04) **Exercise 2.4** Using eqns (2.34)-(2.36), derive the low-temperature expansions

$$\begin{aligned} F(\beta) &= -\frac{1}{\beta} \log\left(\frac{|B|}{\beta w}\right) + \Theta(e^{-\beta wd}) \\ U(\beta) &= \frac{1}{\beta} + \Theta(e^{-\beta wd}) \\ S(\beta) &= \log\left(\frac{|B|e}{\beta w}\right) + \Theta(e^{-\beta wd}) \end{aligned}$$

In this case $\mathcal{X}$ is continuous, and the energy has no gap. Nevertheless, these results can be understood as follows: at low temperature, the molecule is confined to a layer of height of order $1/(\beta w)$ above the bottom of the bottle. It therefore occupies a volume of size $|B|/(\beta w)$. Its entropy is approximately given by the logarithm of such a volume.

3. (IPC: 2.05) **Exercise 2.5** Let us reconsider the above example and assume the bottle to have a different shape, for instance a sphere of radius R . In this case it is difficult to compute explicit expressions for the thermodynamic potentials, but one can easily compute the low-temperature expansions. For the entropy, one gets at large β

$$S(\beta) = \log\left(\frac{2\pi e^2 R}{\beta^2 w^2}\right) + \Theta\left(\frac{1}{\beta}\right)$$

The reader should try to understand the difference between this result and eqn (2.39) and provide an intuitive explanation, as in the previous example. Physicists say that the low-temperature thermodynamic potentials reveal the ‘low-energy structure’ of the system.